

PLI is necessary below density zero: a counterexample to the wPLI extension

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Status and Source Question

Status: counterexample likely valid.

Balcerzak–Poplawski–Wachowicz study, for an ideal $\mathcal{I}$ on $\mathbb{N}$ and a real series $\sum_n x_n$, the set

$$A(\mathcal{I}, (x_n)) = \left\{ t \in \{0, 1\}^{\mathbb{N}} : \sum_n t(n)x_n \text{ is } \mathcal{I}\text{-convergent} \right\},$$

equipped with the fair Bernoulli product measure λ [1]. Their Question 4.6 asks:

Does the statement of Theorem 4.3 remain true if PLI is replaced by wPLI? In particular, does it work for the ideal from Example 3?

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Question 4.6. *Does the statement of Theorem 4.3 remains true if we replace PLI by wPLI for an ideal $\mathcal{I}$? In particular, does it work for the ideal $\mathcal{I}$ from Example 3?*

The answer is negative even for the intended core assertion of Theorem 4.3, namely the existence of a divergent real series with $\lambda(A(\mathcal{I}, (x_n))) = 1$. We prove a converse to the source theorem for all ideals contained in the density-zero ideal.

Definitions

For a real sequence (z_n) and an ideal $\mathcal{I}$ on $\mathbb{N}$, write $z_n \rightarrow_{\mathcal{I}} z$ when

$$\{n : |z_n - z| > \varepsilon\} \in \mathcal{I} \quad (\varepsilon > 0).$$

A series is $\mathcal{I}$ -convergent when its sequence of partial sums is $\mathcal{I}$ -convergent.

The asymptotic-density-zero ideal is

$$\mathcal{I}_d = \left\{ B \subseteq \mathbb{N} : \lim_{N \rightarrow \infty} \frac{|B \cap [1, N]|}{N} = 0 \right\}.$$

Thus $\mathcal{I}_d$ -convergence is statistical convergence.

An ideal $\mathcal{I}$ has the *property of long intervals* (PLI) if some $E \in \mathcal{I}$ contains intervals of every finite length. It has the weak property of long intervals (wPLI) if, for every fixed length r , some $E_r \in \mathcal{I}$ contains infinitely many pairwise disjoint intervals of length r .

The source's ideal $\mathcal{Lac}$ is the smallest ideal containing every range

$$\{b_j : j \in \mathbb{N}\}, \quad b_1 < b_2 < \cdots, \quad b_{j+1} - b_j \longrightarrow \infty.$$

Equivalently, its members are subsets of finite unions of such lacunary ranges. The source proves that $\mathcal{Lac}$ is dense, hence wPLI, but is not PLI [1, Remark 4.4(ii) and Example 3].

Proof Intuition

The missing obstruction is that measure-one random convergence carries much more information than convergence of the original series. Pairing a selector with its complement first shows that the original partial sums are $\mathcal{I}$ -convergent. When $\mathcal{I} \subseteq \mathcal{I}_d$, both the original partial sums and almost every Bernoulli partial-sum path are statistically convergent.

Statistical convergence lets us choose deterministic endpoints at which the random paths converge in probability. After subtracting half of the original partial sum, the increments become independent symmetric variables $\pm x_n/2$. Levy's maximal inequality says that convergence at the selected endpoints controls every partial sum inside the intervening blocks. A single coefficient bounded away from zero would force a fixed-size excursion in its block for every outcome. Hence $x_n \rightarrow 0$.

Now ordinary divergence becomes incompatible with ideal convergence unless the ideal has PLI. Indeed, a partial sum a fixed distance from the ideal limit must, because the terms are small, remain far away for an arbitrarily long interval of preceding indices. One exceptional set in $\mathcal{I}$ therefore contains intervals of all lengths.

Formal Argument

Lemma 1 (Finite Levy maximal inequality). *Let $Y_1, \dots, Y_m$ be independent symmetric real random variables and put $S_k = \sum_{i=1}^k Y_i$. Then, for every $a > 0$,*

$$\mathbb{P}\left(\max_{1 \leq k \leq m} |S_k| > a\right) \leq 2\mathbb{P}(|S_m| > a).$$

Proof. This is the standard reflection argument. On the event that S_k first exits $[-a, a]$, write the remaining sum as Z . Conditionally on the stopped path, Z and $-Z$ have the same law, while the triangle inequality shows that at least one of $|S_k + Z|$ and $|S_k - Z|$ is greater than a . Thus the conditional probability that the terminal sum has absolute value greater than a is at least $1/2$. Summing over the possible first exit times gives the result. $\square$

Lemma 2 (Almost-sure statistical convergence forces null coefficients). *Let $\mathcal{I} \subseteq \mathcal{I}_d$ and let (x_n) be a real sequence. If*

$$\lambda(A(\mathcal{I}, (x_n))) = 1,$$

then $x_n \rightarrow 0$.

Proof. Write $s_n = \sum_{i=1}^n x_i$ and $T_n(t) = \sum_{i=1}^n t(i)x_i$. Let $A = A(\mathcal{I}, (x_n))$ and let $\phi(t) = \mathbf{1} - t$. The map ϕ preserves λ , so $A \cap \phi(A)$ has measure one and is nonempty. Choose t in this intersection. Both $\sum_i t(i)x_i$ and $\sum_i (1 - t(i))x_i$ are $\mathcal{I}$ -convergent. Adding their partial sums shows that

$$s_n \longrightarrow_{\mathcal{I}} s$$

for some $s \in \mathbb{R}$. Since $\mathcal{I} \subseteq \mathcal{I}_d$, (s_n) converges statistically to s .

For almost every selector t , the sequence $T_n(t)$ is likewise statistically convergent; denote its statistical limit by $L(t)$. The limit may be chosen measurable. For example, statistical convergence of $T_n(t)$ implies ordinary convergence of the Cesaro means of the bounded sequence $\arctan(T_n(t))$ to $\arctan(L(t))$, which gives a measurable formula for L on the full-measure convergence set.

Fix $j \geq 1$. Statistical convergence and dominated convergence give

$$\frac{1}{N} \sum_{n=1}^N \lambda(|T_n - L| > 1/j) \longrightarrow 0. \quad (1)$$

Also, the set

$$G_j = \{n : |s_n - s| \leq 1/j\}$$

has asymptotic density one. Consequently, there are arbitrarily large $n \in G_j$ for which

$$\lambda(|T_n - L| > 1/j) < 1/j;$$

otherwise (1) would have positive lower limit. Choose such integers $n_1 < n_2 < \dots$, and put $n_0 = 0$. Then

$$T_{n_j} \longrightarrow L \quad \text{in probability,} \quad s_{n_j} \longrightarrow s.$$

Center the Bernoulli sums:

$$R_n = T_n - \frac{1}{2}s_n = \sum_{i=1}^n \left(t(i) - \frac{1}{2}\right)x_i.$$

The summands are independent and symmetric, taking the values $\pm x_i/2$. We have

$$R_{n_j} \longrightarrow L - \frac{1}{2}s \quad \text{in probability.}$$

Thus the block increments

$$Z_j = R_{n_j} - R_{n_{j-1}}$$

converge to zero in probability. Applying Lemma 1 to the j th block yields, for every $a > 0$,

$$\mathbb{P}\left(\max_{n_{j-1} < k \leq n_j} |R_k - R_{n_{j-1}}| > a\right) \leq 2\mathbb{P}(|Z_j| > a) \longrightarrow 0. \quad (2)$$

If some $i \in (n_{j-1}, n_j]$ satisfied $|x_i| > 4a$, then the corresponding centered increment would have deterministic magnitude $|x_i|/2 > 2a$. The partial sums immediately before and after that increment cannot both have absolute value at most a . Therefore the event on the left of (2) would have probability one. For all sufficiently large j this contradicts (2). Hence

$$\max_{n_{j-1} < i \leq n_j} |x_i| \leq 4a$$

eventually. Since $a > 0$ was arbitrary and the blocks cover a tail of $\mathbb{N}$, $x_n \rightarrow 0$. $\square$

Lemma 3 (Divergent ideal convergence with null terms forces PLI). *Suppose the real series $\sum_n x_n$ is $\mathcal{I}$ -convergent but not ordinarily convergent, and $x_n \rightarrow 0$. Then $\mathcal{I}$ has PLI.*

Proof. Let $s_n = \sum_{i=1}^n x_i$ and $s_n \rightarrow_{\mathcal{I}} s$. Since (s_n) does not converge ordinarily to s , there are $\delta > 0$ and arbitrarily large n such that $|s_n - s| > \delta$. Put

$$E = \{m : |s_m - s| > \delta/2\} \in \mathcal{I}.$$

Fix a positive integer r . Choose one of the above indices n so large that $n \geq r$ and

$$|x_i| < \frac{\delta}{2r} \quad (i \geq n - r + 1).$$

For $m \in \{n - r + 1, \dots, n\}$,

$$|s_m - s| \geq |s_n - s| - \sum_{i=m+1}^n |x_i| > \delta - \frac{r-1}{2r}\delta > \frac{\delta}{2}.$$

Thus E contains an interval of length r . As r was arbitrary, E is a PLI witness. $\square$

Theorem 1 (PLI necessity below density zero). *Let $\mathcal{I} \subseteq \mathcal{I}_d$. If there is a divergent real series $\sum_n x_n$ such that*

$$\lambda(A(\mathcal{I}, (x_n))) = 1,$$

then $\mathcal{I}$ has PLI.

Proof. By the complement-selector argument in Lemma 2, the original series is $\mathcal{I}$ -convergent. The same lemma gives $x_n \rightarrow 0$. Lemma 3 now yields PLI. $\square$

Corollary 1 (Classification below density zero). *Let $\mathcal{I} \subseteq \mathcal{I}_d$. The following are equivalent:*

1. $\mathcal{I}$ has PLI;
2. there is a divergent real series $\sum_n x_n$ for which $\lambda(A(\mathcal{I}, (x_n))) = 1$.

Proof. Theorem 1 proves $(2) \Rightarrow (1)$. The implication $(1) \Rightarrow (2)$ is the core conclusion of source Theorem 4.3; its explicit block construction uses 2^k terms 2^{-k} followed by 2^k terms -2^{-k} inside each PLI interval and applies Chebyshev, Borel–Cantelli, and the Cauchy criterion [1, Theorem 4.3]. $\square$

The wPLI Counterexample

Lemma 4. $\mathcal{Lac} \subseteq \mathcal{I}_d$.

Proof. Let $B = \{b_1 < b_2 < \dots\}$ with $b_{j+1} - b_j \rightarrow \infty$. Given $q \geq 1$, all sufficiently late gaps in B are at least q . Hence

$$|B \cap [1, N]| \leq C_q + \frac{N}{q}$$

for a constant C_q . Its upper asymptotic density is at most $1/q$, and therefore is zero. Finite unions of such ranges also have density zero, as do their subsets. These are exactly the members of $\mathcal{Lac}$. $\square$

Theorem 2 (Negative answer to Question 4.6). *The wPLI hypothesis does not suffice in Theorem 4.3 of the source paper. More precisely, the ideal $\mathcal{Lac}$ from source Example 3 has wPLI, but there is no divergent real series $\sum_n x_n$ satisfying*

$$\lambda(A(\mathcal{Lac}, (x_n))) = 1.$$

Proof. The source proves that $\mathcal{Lac}$ is dense and consequently has wPLI, while its Example 3 proves that $\mathcal{Lac}$ does not have PLI. By Lemma 4, $\mathcal{Lac} \subseteq \mathcal{I}_d$. If the displayed series existed, Theorem 1 would force $\mathcal{Lac}$ to have PLI, a contradiction. $\square$

Printed-Theorem Erratum

Theorem 4.3 additionally prints the conclusion $x_n \not\rightarrow 0$. The displayed construction in its proof actually uses coefficients $\pm 2^{-k}$ on the k th active block, so its coefficients tend to zero. This is not merely a missing observation: Lemma 2 proves that whenever $\mathcal{I} \subseteq \mathcal{I}_d$ and $\lambda(A(\mathcal{I}, (x_n))) = 1$, one necessarily has $x_n \rightarrow 0$. In particular, the extra printed conclusion is impossible for $\mathcal{I} = \mathcal{I}_d$, even though $\mathcal{I}_d$ has PLI. The core measure-one construction used in Corollary 1 is unaffected.

The same mismatch appears in the source’s concluding claim that its constructed series has $\sum_n x_n^2 = \infty$. In fact the k th active block contributes

$$2^{k+1}(2^{-k})^2 = 2^{1-k},$$

so the sum of squares is finite. These source issues are logically separate from Theorem 2, which disproves the intended wPLI extension itself.

Verification and Novelty Check

The proof is non-computational. Its only probabilistic input, Lemma 1, is included with the standard reflection proof. The two deterministic checks are explicit: every $\mathcal{Lac}$ member has density zero, and a divergent ideal-convergent null-term series creates arbitrarily long intervals in one exceptional set.

The run indexes were searched for arXiv:1803.03699, ideal-convergent subseries, PLI, wPLI, the lacunary ideal, and measure-one Bernoulli subseries; no duplicate was found. A bounded web search on July 19, 2026 used the exact question, the exact paper title, the DOI, and the phrases “weak property of long intervals” and “property of long intervals.” Semantic Scholar listed six citing records. The accessible citing sources—including *Another characterization of meager ideals*, *Convergent subseries of divergent series*, and *Ideal boundedness of subseries and rearrangements in Banach spaces vs Banach spaces possessing a copy of c_0* —cite the source only for adjacent category, summable-ideal, or boundedness results; none states or proves an answer to Question 4.6. No later answer was found. This supports, but cannot certify, novelty.

Human-review recommendation: check first the selection of (n_j) in Lemma 2 and the blockwise application of Levy’s inequality. Those are the only steps where measure and statistical convergence interact.

References

- [1] M. Balcerzak, M. Poplawski, and A. Wachowicz, *Ideal convergent subseries in Banach spaces*, Quaestiones Mathematicae 42 (2019), no. 6, 765–779; arXiv:1803.03699. DOI: 10.2989/16073606.2018.1497724.